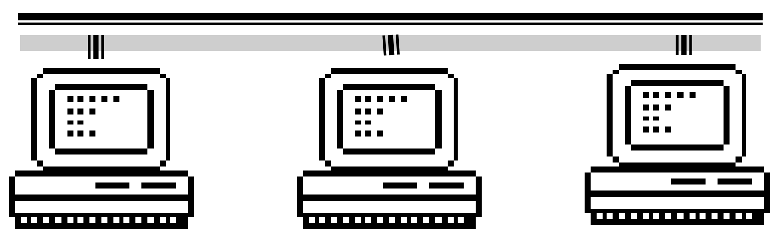
Database system architecture

Chapter 2



What will you learn ?

1. Features of Current database applications
2. Hot topics in database field
3. Database system architecture
   - **From DBMS**
   - **From end user**
4. The Database Administrator
5. DBMS

ANSI/SPARC Architecture

1 features of Current database applications

Integration of the DBMS into the Web environment

- Treat the Web as a database application platform
- Different **integration tools**, including CGI(Common Gateway Interface), Java, scripting languages, Active Server pages, and Oracle's Universal data Server

2 Hot topics in database field

- Data Warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data in support of management's decision-making process
- Store decision-support data (e.g. customers, products, and sales) rather than application-oriented data (e.g. customer invoicing, stock control and product sales)
- Data is not updated in real-time, and usually added as an addition rather than a replacement

- OLAP i.e. Online Analytical Processing is the dynamic synthesis, analysis, and consolidation of large volumes of multi-dimensional data, e.g. City, Property type, and Time as three dimensions in a cube

- Data Mining is the process of extracting valid, previously unknown, comprehensible, and actionable information from large databases and using it to make crucial business decisions.

3. Database system architecture ----from DBMS

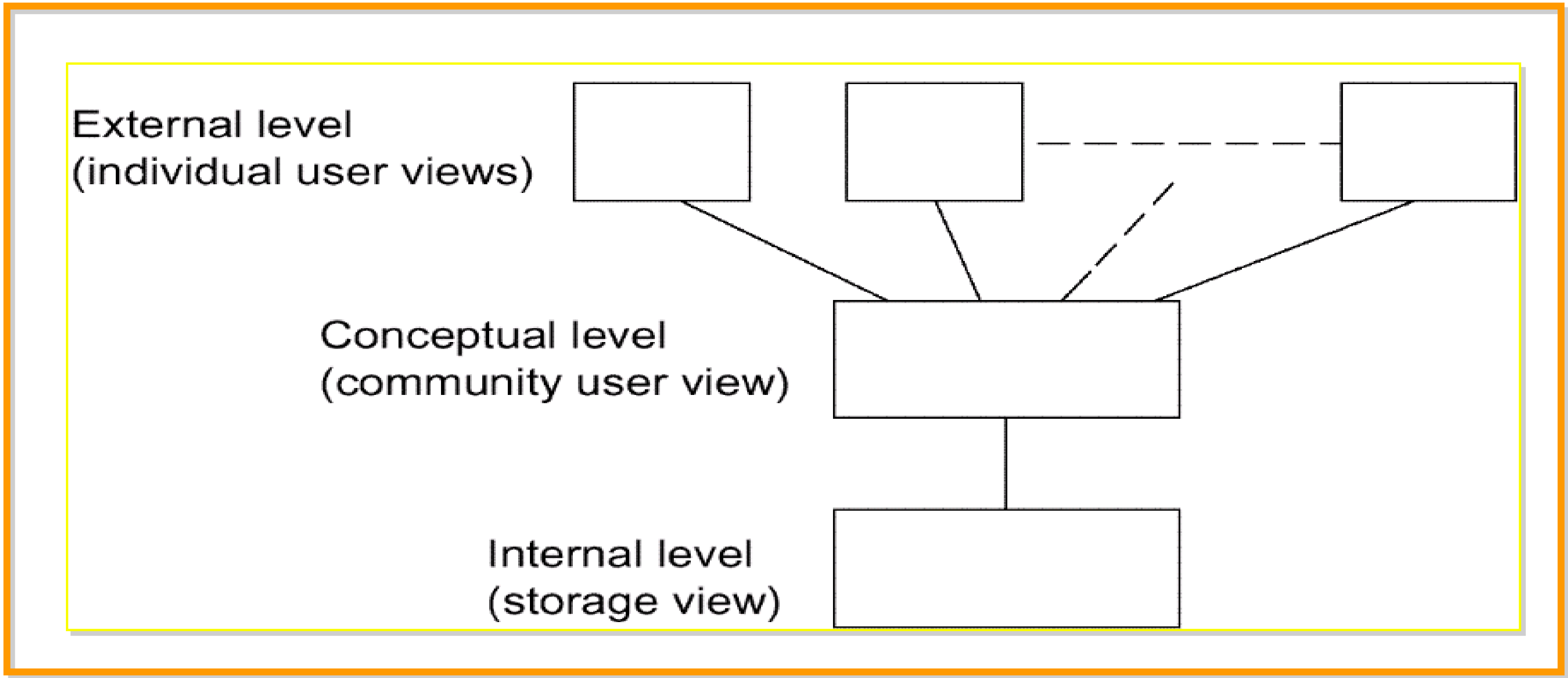


Fig. 2.1 The three levels of the architecture

The Three Levels of the Architecture ANSI/SPARC

(1975) by American National Standard Institute - Standards Planning and requirements Committee(ANSI/SPARC)

internal level (Physical level) :

the way the data is physically stored

external level (user logical level):

the way the data is seen by individual users (application programmer or end user)

conceptual level (community logical level) :

a level of indirection between the other two

An example of the three levels

External (PL/I)

```
DCL 1 EMPP,
      2 EMP# CHAR(6),
      2 SAL FIXED BIN(31);
```

External (COBOL)

```
01 EMPC.
   02 EMPNO PIC X(6).
   02 DEPTNO PIC X(4).
```

Conceptual

```
EMPLOYEE
   EMPLOYEE_NUMBER     CHARACTER (6)
   DEPARTMENT_NUMBER   CHARACTER (4)
   SALARY              NUMERIC (5)
```

Internal

```
STORED_EMP    BYTES=20
   PREFIX     TYPE=BYTE(6),OFFSET=0
   EMP#       TYPE=BYTE(6),OFFSET=6,INDEX=EMPX
   DEPT#      TYPE=BYTE(4),OFFSET=12
   PAY        TYPE=FULLWORD,OFFSET=16
```

Fig. 2.2 An example of the three levels

Conceptual level

The conceptual view is a representation of the entire information content of the database.

The conceptual view is defined by means of the conceptual schema, which includes the definitions of each of the various conceptual records types

The "conceptual schema" is really little more than a simple union of all of the individual external schemas, plus certain security and integrity constraints.

E.g. conceptual schema of Database "Order"

Table:

Customer(Cno,Company,City,Tel)

Order(Ono,Odate,Freight)

Order_item(Ono,Pno,Quantity)

Product(Pno,Pname,Price)

integrity constraints

security constraints

The conceptual schema is written using the conceptual DDL---just define information content

The External Level

External view is the individual user view.

User and language

- application programmer:
  - programming language ,e.g., PL/I, C++, Java
  - Proprietary language-4GLs
- end user:
  - query language
  - Special-purpose language (supported by application program)

all such languages include a data sublanguage,it include DDL and DML.

第四代语言（4GL）是一个简洁的、高效的非过程编程语言，用来提高DBMS的效率。在第四代语言中，用户定义“做什么”而不是“如何做”。第四代语言依靠更高级的第四代工具，用户可以使用这个工具定义参数来生成应用程序。第四代语言内嵌了如下组件：

查询语言

报表生成器

电子数据表

数据库语言

应用程序生成器

生成应用程序的高级语言

结构化查询语言（SQL）

通过例子查询（QBE）

第四代语言又称为高生产率语言

确定一个语言是否是一个4GL，主要应从以下标准来进行考察：

（1）生产率标准：4GL一出现，就是以大幅度提高软件生产率为己任的，4GL应比3GL提高生产率一个数量级以上。

（2）非过程化标准：4GL基本上应该是面向问题的，即只需告知计算机“做什么”，而不必告知计算机“怎么做”。当然4GL为了适应复杂的应用，而这些应用是无法“非过程化”的，就允许保留过程化的语言成分，但非过程化应是4GL的主要特色。

（3）用户界面标准：4GL应具有良好的用户界面，应该简单、易学、易掌握，使用方便、灵活。

（4）功能标准：4GL要具有生命力，不能适用范围太窄，在某一范围内应具有通用性。

4GL存在着以下严重不足：

（1）4GL虽然功能强大，但在其整体能力上却与3GL有一定的差距。这一方面是语言抽象级别提高以后不可避免地带来的（正如高级语言不能做某些汇编语言做的事情）；另一方面是人为带来的，许多4GL只面向专项应用。有的4GL为了提高对问题的表达能力，提供了同3GL的接口，以弥补其能力上的不足。如Oracle提供了可将SQL语句嵌入C程序中的工具PRO*C。

（2）4GL由于其抽象级别较高的原因，不可避免地带来系统开销庞大，运行效率低下（正如高级语言运行效率没有汇编语言高一样），对软硬件资源消耗严重，应用受硬件限制。

（3）由于缺乏统一的工业标准，4GL产品花样繁多，用户界面差异很大，与具体的机器联系紧密，语言的独立性较差（SQL稍好），影响了应用软件的移植与推广。

（4）4GL主要面向基于数据库应用的领域，不宜于科学计算、高速的实时系统和系统软件开发。

Data sublanguage (DSL)

Data Definition Language (DDL): supports the definition or declaration of database objects.

Data Manipulation Language (DML): supports the manipulation or processing of database objects.

- Data sublanguage is embedded within the corresponding host language.

Data sublanguage is responsible for database operations, host language for non database facilities(e.g. user interface, data input and output, computational operations).

SQL

- Tightly coupling ⇔ loosely coupling

External view is defined by means of an external schema. It is written using the external DDL.

The Internal Level

The internal view is a low-level representation of the entire database.

internal view=storage structure or stored database

Internal level =physical level?

The internal view is described by means of the internal schema.

The internal schema is written using the internal DDL.

Application operates directly at internal level, recommended?

How the three levels of the architecture are typically realized in a relational system?

The conceptual level in such a system will definitely be relational:

- relational tables
- relational operators→result:table

A given external view will typically be relational:

- Table or View

the internal level will not be relational:

- stored records, pointers, indexes, hashes, etc.

Detailed system architecture

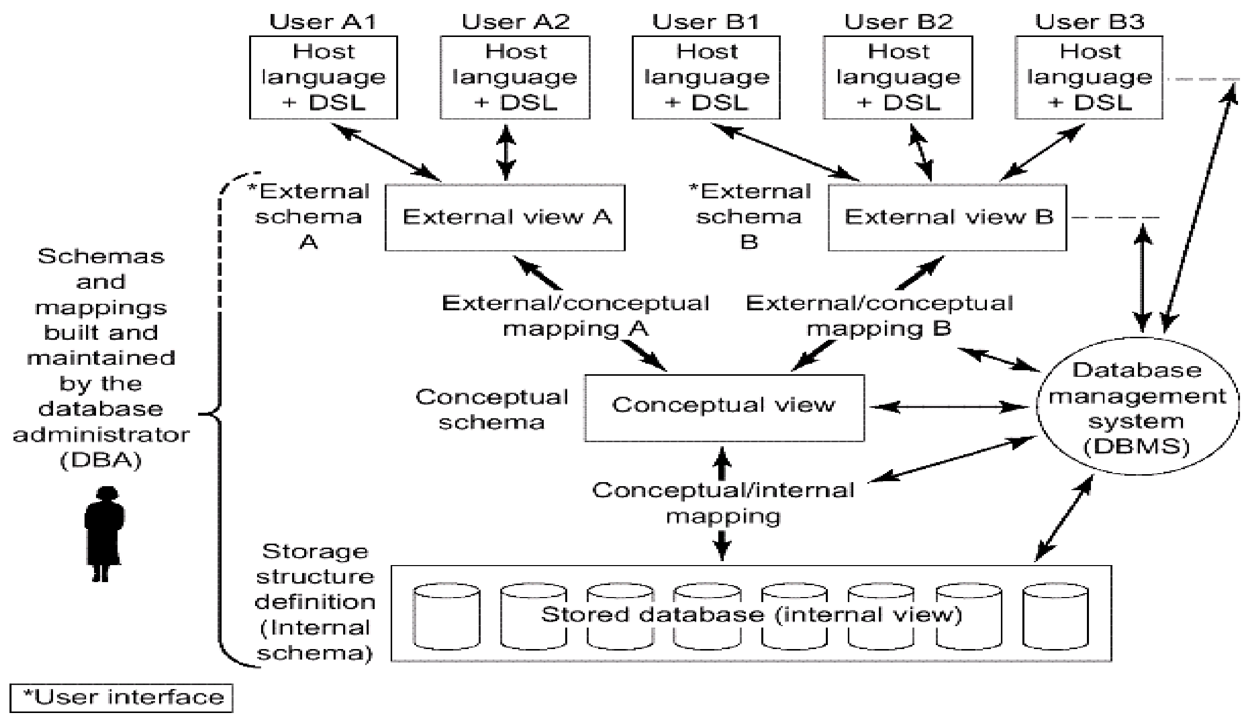


Fig. 2.3 Detailed system architecture

Mappings

conceptual/internal mapping:

- Defines the correspondence between the conceptual view and the stored database;
- It specifies how conceptual records and fields are represented at the internal level.
- It is the key to physical data independence :If the structure of the stored database is changed ,the conceptual/internal mapping must be changed accordingly, so that the conceptual schema can remain invariant.

Mappings

external/conceptual mapping

Defines the correspondence between a particular external view and the conceptual view.

Fields can have different data types; field and record names can be changed; several conceptual fields can be combined into a single (virtual) external field;

Any number of external views can exist at the same time;

Any number of users can share a given external view;

Different external views can overlap;

It is the key to logical data independence.

Mappings

external/external mapping

Most systems support external/external mapping, rather than always requiring an explicit definition of the mapping to the conceptual level.

Mapping and Data Independence

data independence: the ability to modify a schema definition in one level without affecting a schema definition in the next higher level

Logical data independence: the capacity to change the conceptual schema without having to change external schemas or application programs.

Physical data independence: the capacity to change the internal schema without having to change the conceptual (or external) schemas.

The mapping of three-schema architecture can make it easier to have true data independence.

. Database system architecture(continued) ----from end users

single user

主从

Client/Server Architecture

Distributed Processing

Single user

The whole database system, including application, DBMS and data is in one computer, used by just one user. No data share

主从式结构

一个主机带多个终端

application, DBMS,data 放在主机上　所有处理任务有主机完成

各个用户通过主机的终端并发地存取数据库，共享数据资源

优点：

简单 数据易于管理和维护

缺点：

终端用户数目增加到一定程度，主机任务会过分繁重，成为瓶颈，系统性能大幅度下降

系统可靠性不高

Client/Server Architecture

Server:

supports all of the basic DBMS functions.

Client:

the various applications that run on top of the DBMS—both user-written applications and built-in applications, i.e., applications provided by the DBMS vendor or by some third party.

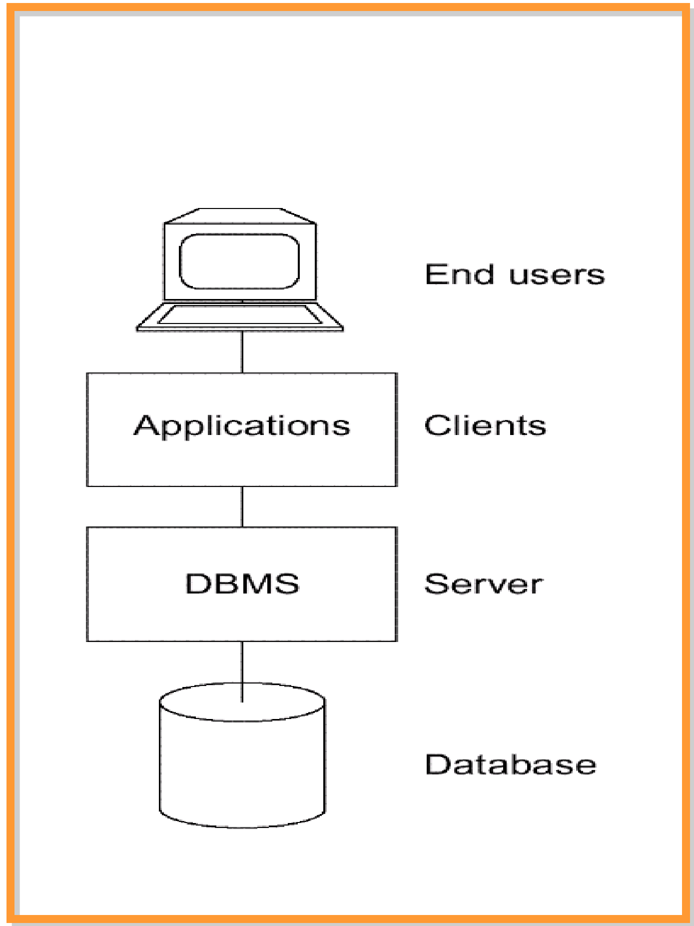


Fig. 2.5 Client/server architecture

Client/Server Architecture(Cont.)

User-written applications:

regular application programs

written in C ,COBOL ,Java,…

Vendor-provided applications(tools):

assist in the creation and execution of other applications

E.g. *report writer*:

allow user to obtain formatted reports through *report writer language.*

Vendor-provided tools

- Query language processors;
- Report writers;
- Business graphics subsystems;
- Spreadsheets;
- Natural language processors;
- Statistical packages;
- Copy management or "data extract" tools;
- Application generators (including 4GL processors);
- Other application development tools, including computer-aided software engineering (CASE) products;

Utilities

are programs designed to help the DBA with various administration tasks

--operate at the external level, perhaps provided by some third party

--operate at the internal level, provided by DBMS vendor

- Load routines
- Unload/reload routines
- Reorganization routines
- Statistical routines
- Analysis routines

Benefit

Server (database) and client (application) processing are being done in **parallel**. **Response time** and **throughput** should thus be improved.

The server machine might be a custom-built machine that is tailored to the DBMS function (a "database machine") and might thus provide better DBMS performance.

The client machine might be a personal workstation, tailored to the needs of the end user and thus able to provide better interfaces, faster responses, and overall improved ease of use to the user.

Benefit(continued)

Several different client machines might be able (in fact, typically will be able) to access the same server machine. Thus, a single database might be shared across several distinct client systems

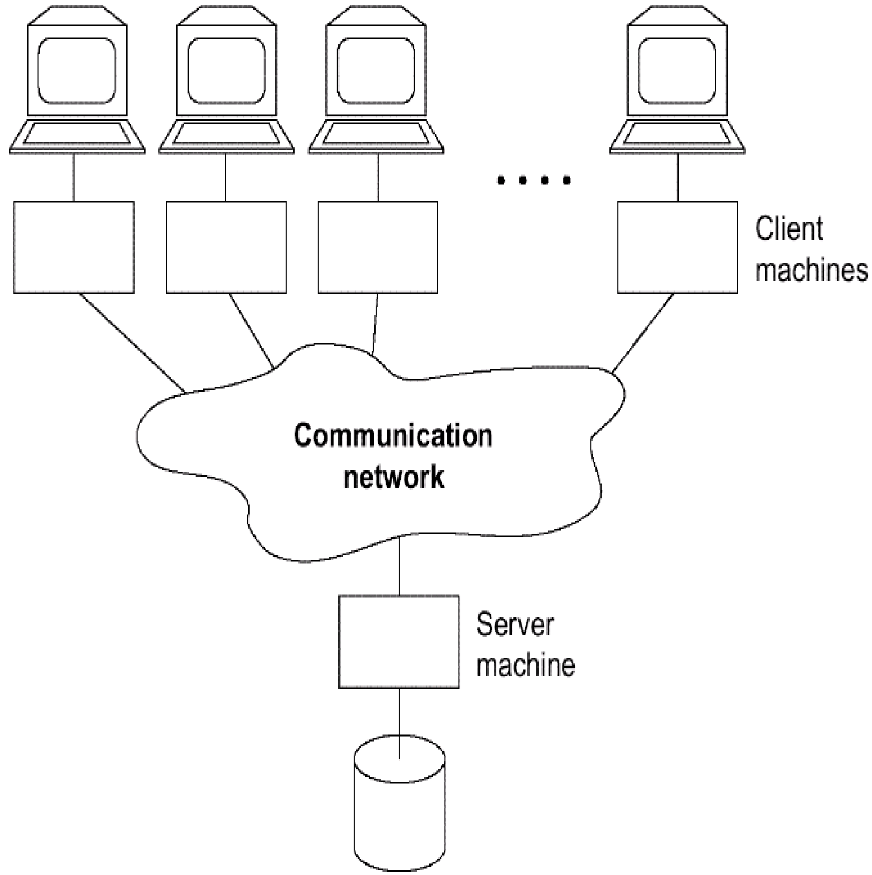


Fig. 2.7 one server machine, many client machines

优点

减少了网络上的数据传输量，提高了系统的性能、吞吐量和负载能力

数据库更加开放。

客户机和服务器能在多种不同的硬件和软件平台上运行，可使用不同厂商的数据库应用开发工具，应用程序具有更强的可移植性，可减少软件维护开销

缺点

网络中有一台数据库服务器时，为众多的客户服务，容易成为瓶颈，制约系统性能

有多台数据库服务器时，数据处理、管理与维护有困难。

distributed database system

The client might be able to access many servers simultaneously (i.e., a single database request might be able to combine data from several servers). In this case, the servers look to the client—from a logical point of view—as if they were really a single server, and the user does not have to know which machines hold which pieces of data.

优点：

适应了地理上分散的组织对于数据库应用的需求

缺点：

数据分布存放给数据的处理、管理与维护带来了困难

系统效率会受到网络交通的制约

Each machine runs both client (s) and server

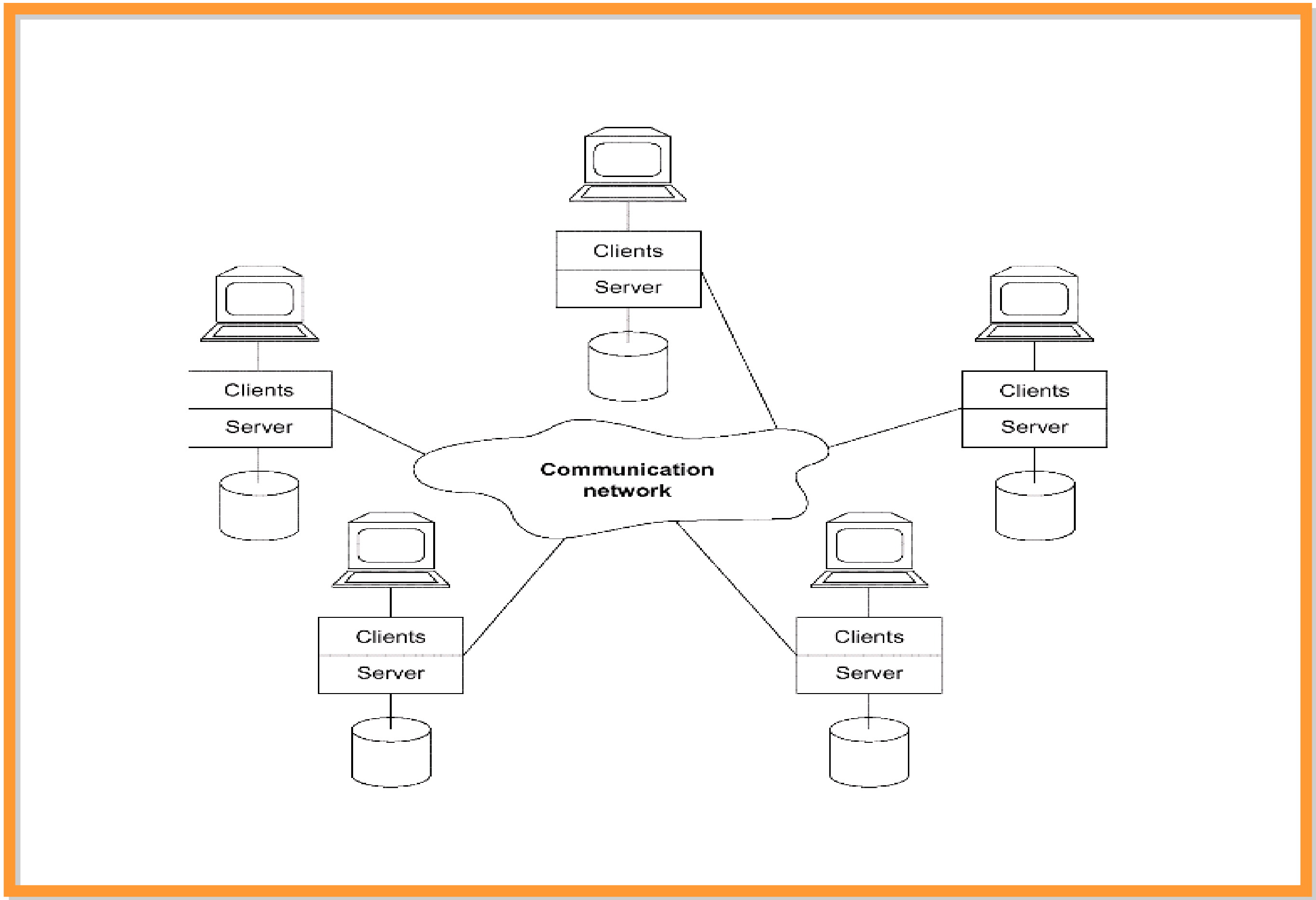


Fig. 2.8 Each machine runs both client (s) and server

4.The Database Administrator

DA(data administrator) is the person who *makes the strategic and policy decisions* regarding the data of the enterprise.

DBA is the person who provides the necessary technical support for *implementing those decisions*.

Tasks of DBA

- Defining the conceptual schema(using conceptual DDL)
  - DA :decide the content of the database at an abstract level (Logical DB design)
  - DBA create the corresponding conceptual schema
  - object form and source form of schema
  - SQL: create table
- Defining the internal schema (using internal DDL)
  - The DBA must also decide how the data is to be represented in the stored database (physical database design).
  - DBA create the corresponding internal schema
  - Define the associated mapping
  - Internal schema and mapping exists in both object and source form
- Liaising with users
  - Liaise with users to ensure that the data they need is available.
  - Consulting on application design, providing technical education ,assisting with problem determination an resolution.
  - write the necessary external schemas and the corresponding external/conceptual mappings using external DDL .(SQL: create view)
  - external schema and mapping exists in both object and source form

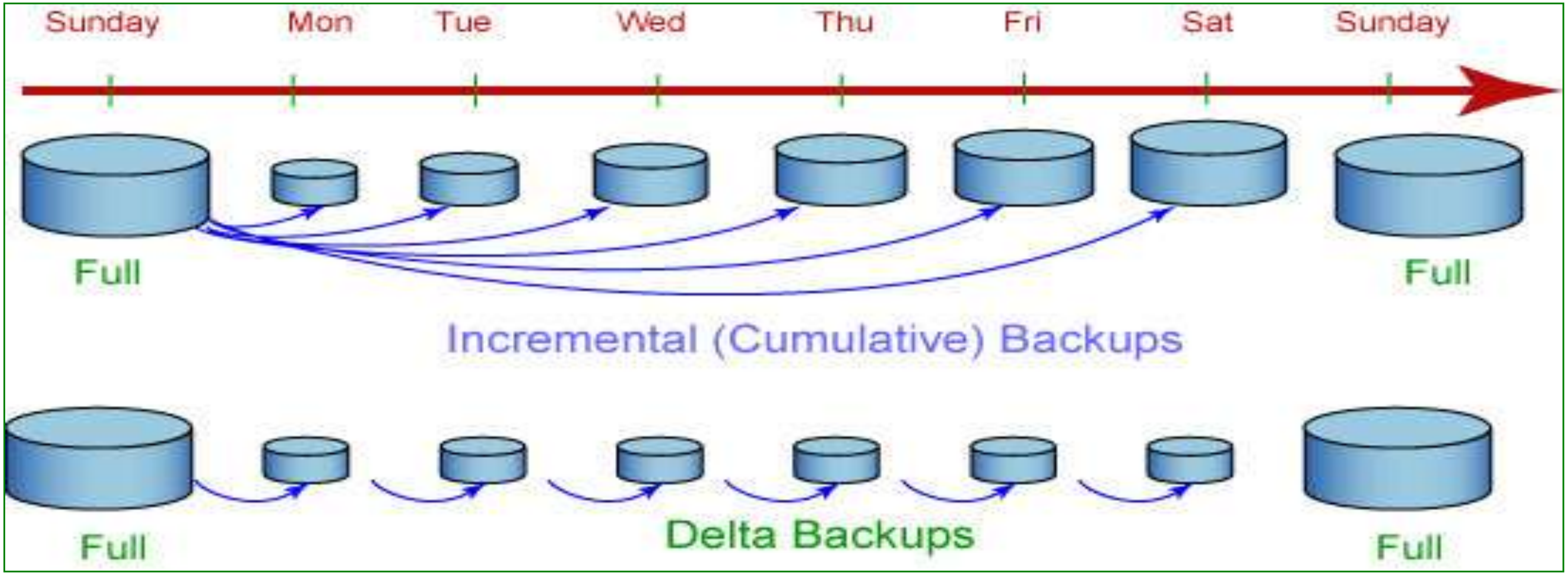
Tasks of DBA

- Defining security and integrity constraints
- Defining dump and reload policies
  - DBA define and implement an damage control scheme involving
    - Periodic unloading or dumping of the database to backup storage
    - Reloading the database when necessary from the most recent dump
- Monitoring performance and responding to changing requirements

数据转储(备份)

增量：DB2 备份自上次完全数据库备份以来所更改的所有数据。

delta：DB2 将只备份自上一次成功的完全、增量或差异备份以来所更改的数据。



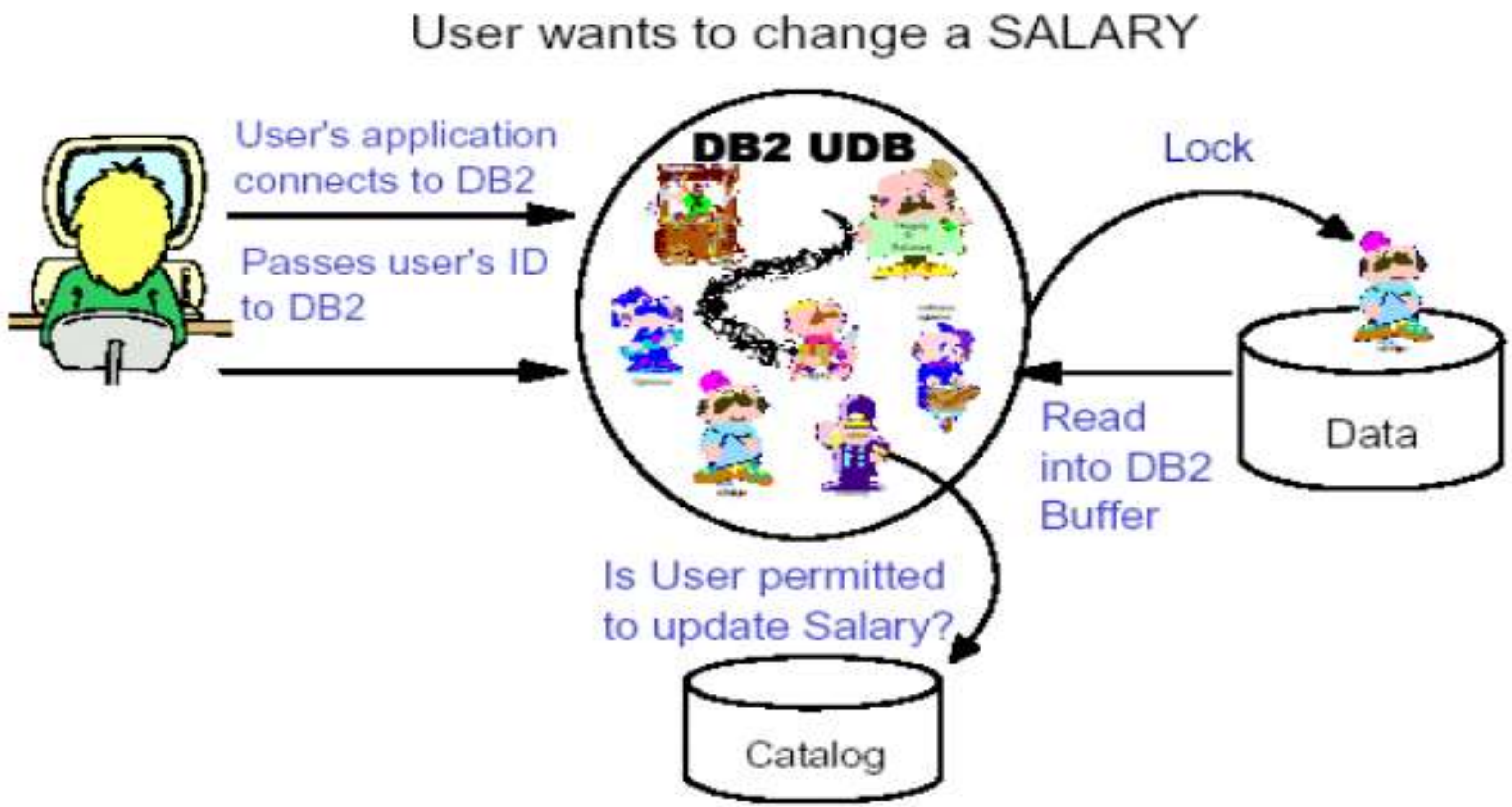
- 如果星期五进行增量备份之后发生崩溃，那么可以先恢复第一个星期日的完全备份，然后恢复星期五进行的增量备份。
- 如果星期五进行 delta 备份之后发生崩溃，那么可以先恢复第一个星期日的完全备份，然后恢复从星期一到星期五（包括星期五）所进行的每次 delta 备份。

5. DBMS

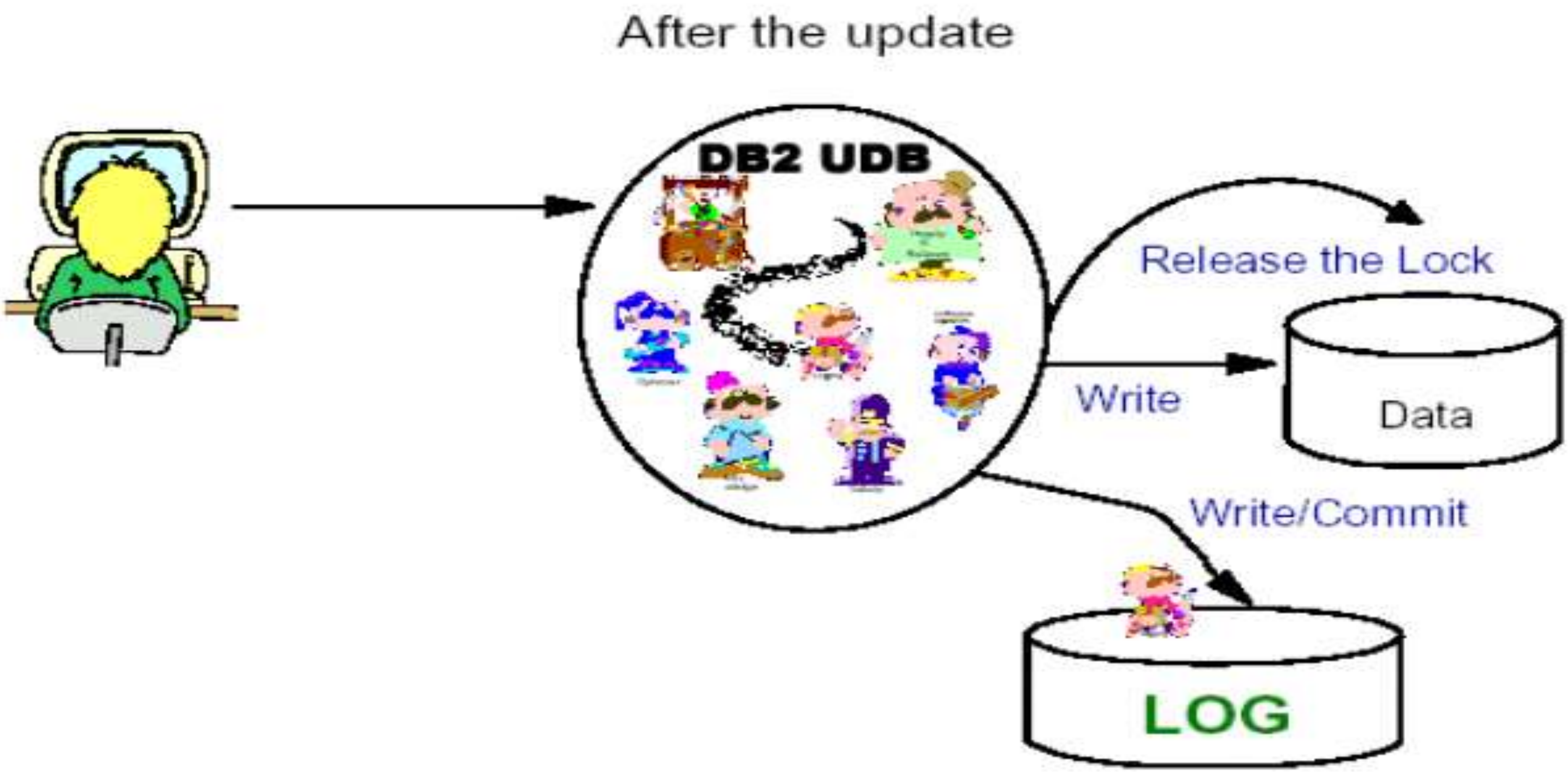
Database management system (DBMS) is the software that handles all access to the database.

- A user issues an access request, using some particular data sublanguage (typically SQL).
- The DBMS intercepts that request and analyzes it.
- The DBMS inspects the external schema for that user, the corresponding external /conceptual mapping, the conceptual schema, the conceptual/internal mapping, and the storage structure definition.
- The DBMS executes the necessary operations on the stored database.

How DB2 Handles an SQL “Change” Request



Finish the change request



Functions of DBMS

Data definitions

DBMS can

accept data definition (external schemas,the conceptual schema,the internal schema and all associated mappings) in source form.

Convert source form schema to the appropriate object form.

Include DDL processor or DDL compiler

Functions of DBMS (Continued)

Data manipulation

DBMS can

handle requests to retrieve, update, delete ,insert data to the database

Include DML processor or DML compiler

DML requests can be

planned request: issued from prewritten application

Characteristic of operational or production applications

unplanned request: issued interactively via some query language processor.

Characteristic of decision support applications

Functions of DBMS (Continued)

Optimization and execution

optimizer, run-time manager

Data security and integrity

Data recovery and concurrency

DBMS and other relative software are called transaction manager or transaction processing monitor

Data dictionary

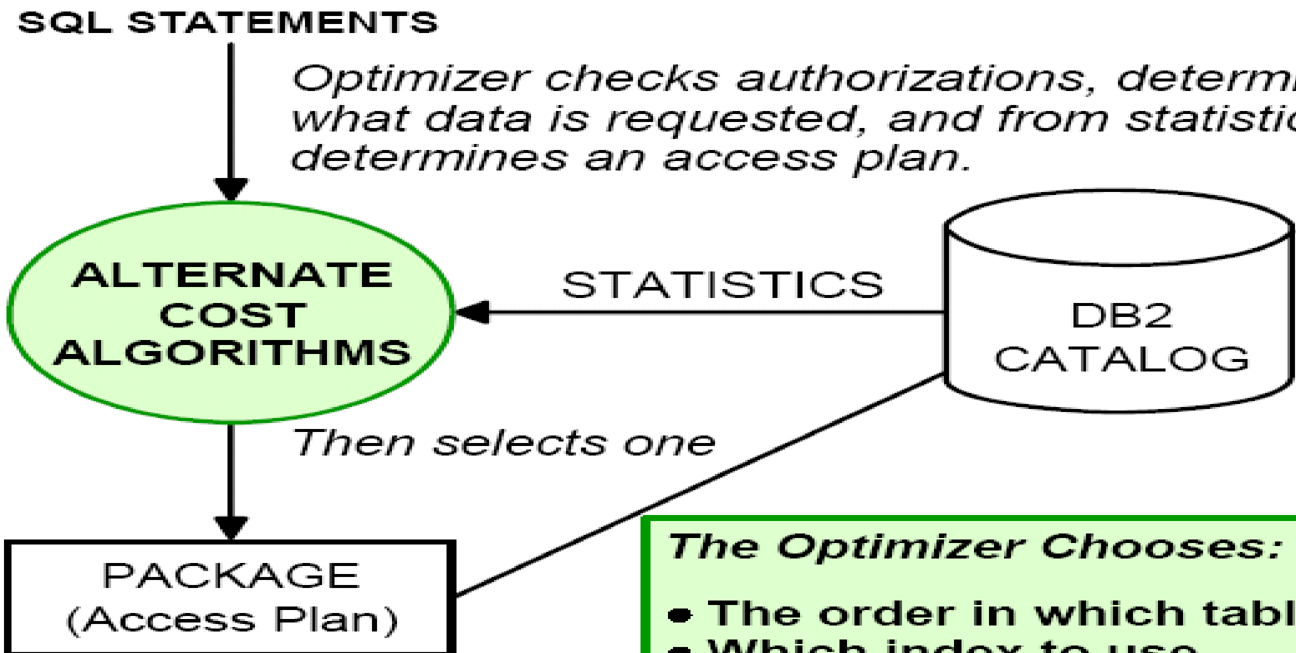
data about data—metadata

store various schemas , mappings, security and integrity constraints in both source and object form.

Performance: efficiently

Functions of DBMS (Continued)

Optimization and execution

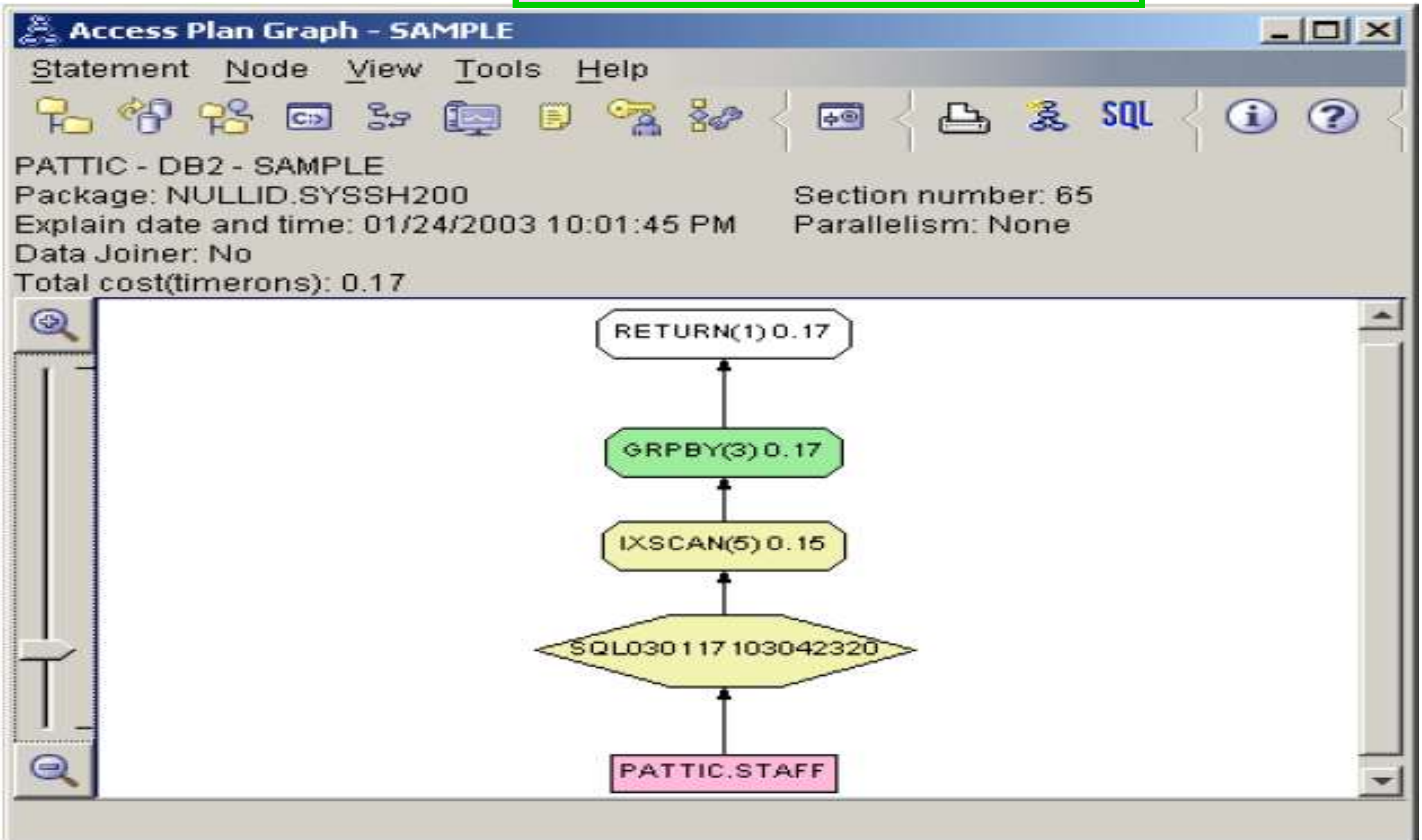


The Optimizer Chooses:

- **The order in which tables are accessed**
- **Which index to use**
- **The join method**
- **The type of scan required**

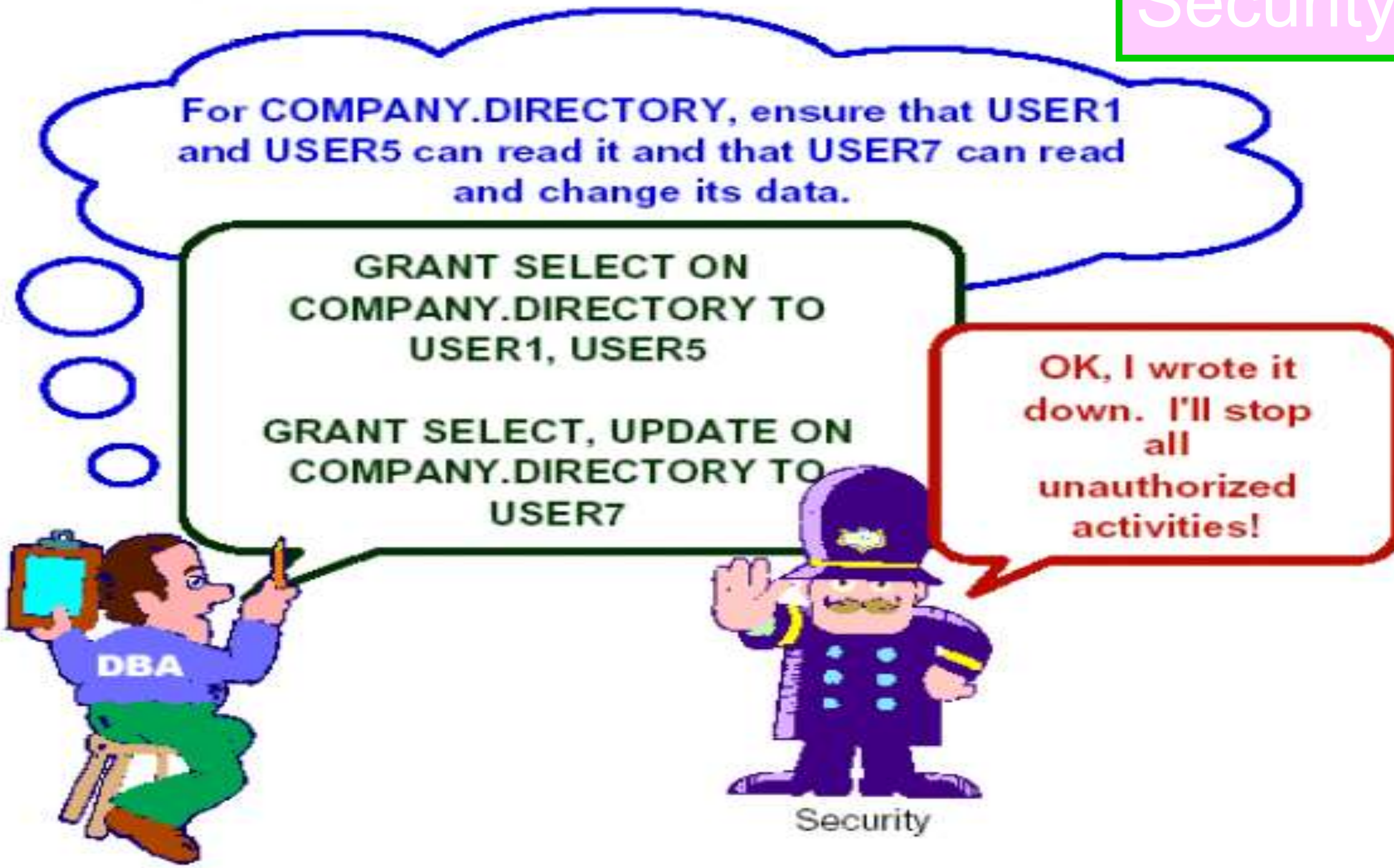
Functions of DBMS (Continued)

Access Plan



Functions of DBMS (Continued)

Security



Functions of DBMS (Continued)

Integrity

LAST NAME	FIRST NAME	MI	DEPT	PHONE
JONES	ANDY	B	SALES	6-3357
JONES	ANN	S	ACC	6-9271
JONES	BRENDA	L	OPER	6-8956
JONES	FRANK	D	ENG	6-7224
JONES	LARRY	R	SALES	6-4367
JONES	LARRY	R	SALES	6-4588
JONES	SANDRA	D	OPER	6-8499
JONES	SANDY		REPRO	6-2875

Functions of DBMS (Continued)

Concurrency

LAST NAME	FIRST NAME	MI	DEPT	PHONE
JONES	**ANDY**	**B**	**SALES**	**6-3357**
JONES	ANN	S	ACC	6-9271
JONES	BRENDA	L	OPER	6-8956
JONES	FRANK	D	ENG	6-7224
JONES	**LARRY**	**R**	**SALES**	**6-4367**
JONES	LARRY	R	SALES	6-4588
JONES	SANDRA	D	OPER	6-8499
JONES	**SANDY**		**REPRO**	**6-2875**

A Page of Data

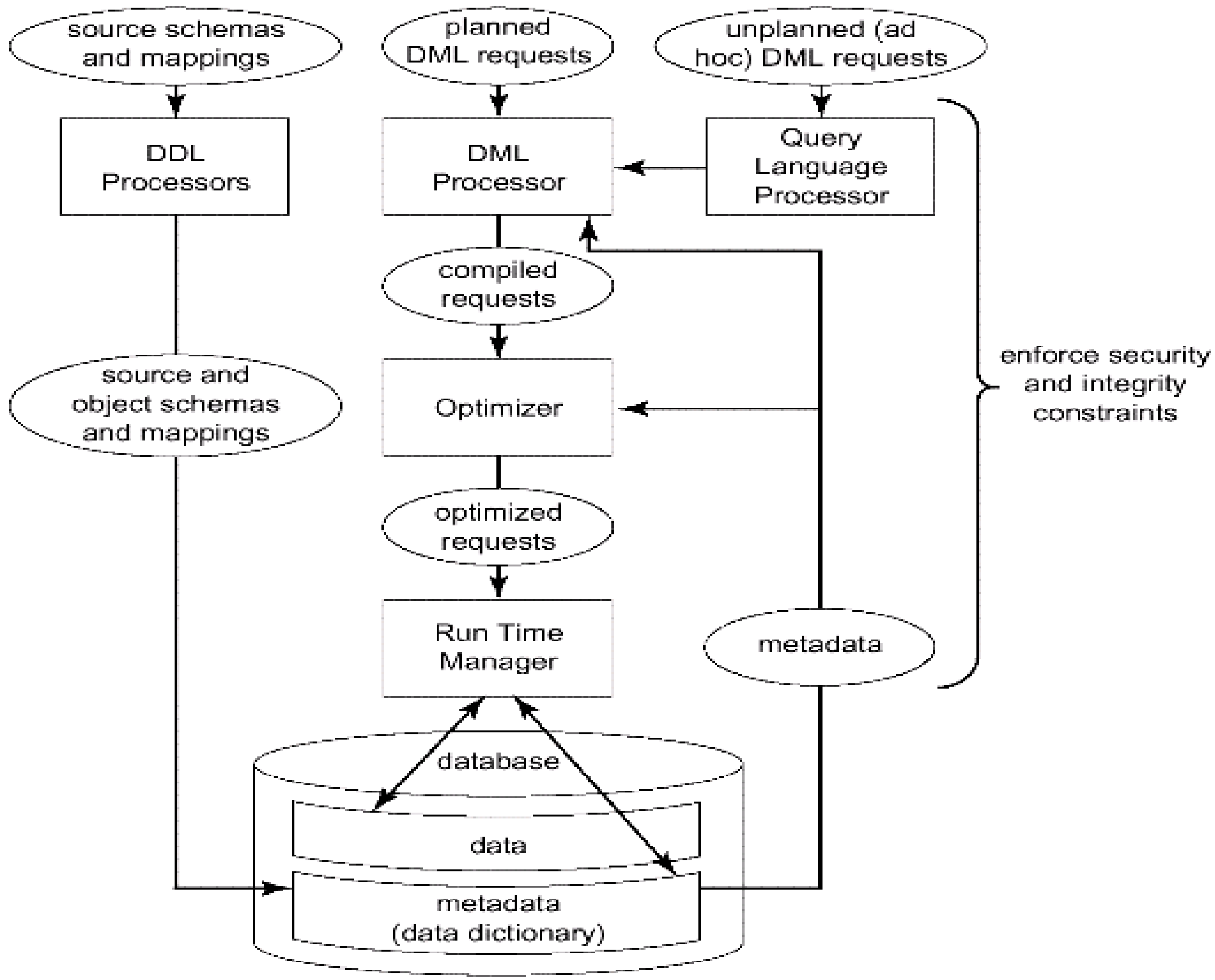


Fig. 2.4 Major DBMS functions and components

- Data definition
- Data manipulation
- Optimization and execution
- Data security and integrity
- Data recovery and concurrency
- Data dictionary

➔The overall purpose of DBMS is to provide the user interface to DB.

Which level?

DBMS & File management system(FMS)

FMS is the component of the underlying operation system that manages stored files----closer to the disk than the DBMS is.

DBMS is built on top of some kind of FMS

Users of FMS can create and destroy stored files and perform simple retrieval and update operations on stored records in such files

In contrast to DBMS…. FMS

- Are not aware of the internal structure of stored records and can not handle requests that rely on a knowledge of that structure
- Provide little or no support for security and integrity constraints.
- Provide little or no support for recovery and concurrency constraints
- No dictionary concept
- Provide much less data independence
- Files are not integrated or shared in the same sense that DB is